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IEEE TWC- Decision on Manuscript ID Paper-TW-Mar-26-1022

1 message

IEEE Transactions on Wireless Communications <onbehalfof@manuscriptcentral.com> Sat, Jun 6, 2026 at 9:46 PM

Reply-To: zhouyong@shanghaitech.edu.cn
To: xw2008@columbia.edu
Cc: yz4891@columbia.edu, xl2427@columbia.edu, xw2008@columbia.edu, eejzhang@ust.hk, jun.jzhang4@gmail.com, zhouyong@shanghaitech.edu.cn, eic.twc@ece.ubc.ca, admin.twc@ece.ubc.ca

06-Jun-2026

*** Please submit your revision in double-column. The detailed format requirements can be found on the TWC website (https://www.comsoc.org/publications/journals/ieee-twc/submit-manuscript).

Dear Prof. Wang:

Manuscript ID Paper-TW-Mar-26-1022, titled "A Semi-amortized Lifted Learning-to-Optimize Masked (SALLO-M) Transformer Model for Scalable and Generalizable Beamforming" and submitted to the IEEE Transactions on Wireless Communications, has been reviewed. The comments of the reviewer(s) are included at the bottom of this letter.

The reviewer(s) have suggested some major revisions to your manuscript. Therefore, I invite you to respond to the reviewer(s)' comments and revise your manuscript.

I have received comments from four reviewers, who raised critical concerns on the novelty, channel models, real-time claim, ablation study, and fair performance comparison.
- A key feature of the proposed SALLO Transformer is its ability to handle varying number of users and antennas. As the reviewers mentioned, the underlying techniques adopted in the paper have been widely used in the literature. It is essential to further elaborate the novelty of the proposed method as well as conduct more comprehensive ablation study (e.g., residual update design, curriculum design, without two-step WMMSE initialization).
- According to the system model, only small-scale fading is considered in the paper, while the large-scale path-loss is ignored. The communication heterogeneity (different users have different communication distances) is necessary to be considered, and may largely affect the scalability and the generalizability of the proposed model. Hence, the authors should account this issue in the revision.
- The authors are required to provide more convincing evidence that the method can truly meet real-time requirement. Based on Table 2, the current inference time may not meet the millisecond requirements at the physical layer. Please clarify the target latency of the typical applications.
- The performance comparison with WMMSE should be clearly elaborated. It is also suggested to compare with multi-start WMMSE.
- The authors are required to account for more practical channel models, including imperfect CSI, spatially correlated/Rician/3GPP-like channels.
- Detailed simulation parameters should be presented.

It should be emphasized that a major revision decision does not imply eventual acceptance. A future version of the paper will only be accepted if the revision addresses the major concerns, and the reply is highly persuasive to the Editors and Reviewers.

Once the revised manuscript and a point-by-point reply to the reviewer comments is prepared, you can upload them both and submit through your Corresponding Author Center. The reply should be designated as a "supplementary file.

When you are ready to submit your revision, visit the following link:
https://ieee.atyponrex.com/submission/submissionBoard/REX-PROD-2-7352D3D4-D1D7-42B4-AD2B-A42AFE12531E-DC58475D-8BF4-43F9-85C1-4903FEB63FC6-61545/current?idtype=external

IMPORTANT: Your original files are available to you when you upload your revised manuscript. Please delete any redundant files before completing the submission.

When you revise your manuscript, please make sure that you comply with our manuscript length requirements. Your revised manuscript should be returned within EIGHT WEEKS. We will consider the manuscript as rejected if the revised manuscript is not received within EIGHT WEEKS from the date of this letter.

Once again, thank you for submitting your manuscript to the IEEE Transactions on Wireless Communications and I look forward to your revision.

Sincerely,

Dr. Yong Zhou
Editor, IEEE Transactions on Wireless Communications
zhouyong@shanghaitech.edu.cn

Reviewer(s)' Comments to Author:
Reviewer: 1

Comments to the Corresponding Author
This manuscript proposes a semi-amortized lifted learning-to-optimize masked Transformer model for downlink beamforming optimization. I have the following comments.
1. A moderate reduction in the number of keywords is suggested.
2. The abbreviation for "Deep learning" has been provided, yet the full term continues to appear multiple times throughout the rest of the text.
3. An unnecessarily large amount of blank space has been left around Fig 3.
4. The journal names in the reference list should be abbreviated.
5. In your sparse channel modeling, the antenna spacing is assumed to be fixed. A problem arises: how can half-wavelength antenna spacing be guaranteed when antenna selection is carried out simultaneously?
6. According to the ablation study, several advanced training methods highlighted in the manuscript do not appear to significantly improve performance.

Reviewer: 2

Comments to the Corresponding Author
This manuscript proposes a semi-amortized lifted learning-to-optimize masked Transformer framework, named SALLO-M, for scalable and generalizable downlink beamforming in MU-MISO systems. The method combines dual user-antenna tokenization, structured masking, gradient-based refinement, and curriculum-based training to handle varying user and antenna configurations. The topic is relevant and the framework is comprehensive, but the novelty, fairness of comparison, robustness validation, and reproducibility need further strengthening.

1. The proposed framework combines several existing techniques, including Transformer, semi-amortized L2O, masking, curriculum learning, and sample replay. The authors should more clearly explain which components are technically novel for beamforming and distinguish the work from closely related Transformer/L2O-based beamforming methods.
2. The claim that the proposed method outperforms WMMSE is strong and requires a fairer comparison. The authors should specify the WMMSE initialization, stopping criterion, iteration budget, and consider stronger WMMSE variants such as multi-start WMMSE.
3. The current evaluation mainly uses i.i.d. Gaussian and simplified sparse channels with perfect CSI. The authors should add experiments under imperfect CSI, spatially correlated/Rician/3GPP-like channels, and train-test channel mismatch to support the claimed generalizability and robustness.
4. The paper should provide more implementation details for all learning-based baselines, including model size, training settings, and hyperparameter tuning. It should also report single-instance latency, FLOPs, memory usage, training cost, and confidence intervals over multiple random seeds.

Reviewer: 3

Comments to the Corresponding Author
This paper proposes a semi-amortized lifted learning-to-optimize masked Transformer framework for scalable and generalizable downlink beamforming in MU-MISO systems. The proposed method combines auxiliary-variable lifting, dual user–antenna tokenization, masking, gradient refinement, and curriculum-based training to improve sum-rate performance across different user–antenna configurations. Nevertheless, the manuscript could be further revised and improved, and there are several concerns as detailed below.
1. Residual connections are an important design component in the proposed SALLO framework, but their effectiveness is mainly motivated by general experience from ResNet/Transformer architectures. The paper should provide a dedicated ablation study to verify whether the residual update design indeed improves convergence or final beamforming performance in this specific problem.
2. The effect of curriculum learning itself is not separately validated. Since curriculum learning, random masking, and multi-configuration training are used together, the authors should include comparisons such as random-order training or non-curriculum multi-configuration training to isolate the actual contribution of the curriculum schedule.
3. WMMSE is a very strong and widely used baseline for linear precoding, and the comparison with WMMSE should be described in more detail. In particular, the authors should clearly specify the initialization, stopping criterion,

maximum number of iterations, and whether multi-start WMMSE is considered.
4. In the sparse overloaded scenario, the proposed method uses a two-step WMMSE initialization, which may affect the fairness of comparison. The authors should explain this design more carefully and provide additional results without the two-step WMMSE initialization in the sparse overloaded regime.
5. It is not clear whether the auxiliary variable C is also masked after each Transformer layer. The paper explicitly states that the beamformer W is masked, but if C is not similarly masked, invalid user or antenna positions may still carry nonzero auxiliary information into later layers.
6. The manuscript would benefit from a careful language and notation check. For example, terms such as “LayerNorm,” “Layer-Norm,” “LN,” and “TN” are used somewhat inconsistently, and they should be unified or clearly defined.

Reviewer: 4

Comments to the Corresponding Author
This paper proposes a semi-amortized lifted learning-to-optimize masked Transformer model for scalable and generalizable downlink beamforming in MU-MISO systems. The proposed SALLO-M framework jointly updates an auxiliary representation and the beamformer across multiple layers, and further refines the output at each layer using projected gradient ascent. The reviewer has several concerns regarding the motivation, novelty, real-time claim, formulation clarity, and simulation validation given as follows:

1. The paper repeatedly motivates the proposed method as a real-time beamforming solution. However, the proposed method remains an iterative method at inference time. The authors should provide more convincing evidence that the method can truly meet real-time requirements, e.g., latency under realistic coherence-time constraints, CPU/GPU deployment results, batch-size-one inference time, and a complexity/latency comparison under the same accuracy level.

2. The related-work discussion mainly focuses on conventional optimization, generic DL beamforming, GNNs, Transformers, and foundation-model-based methods. However, real-time beamforming can also be addressed through predictive beamforming [1], [2]. The authors should clarify how the proposed L2O-based beamforming differs from predictive beamforming in terms of assumptions, latency, robustness, and applicable channel dynamics.

[1] F. Liu, W. Yuan, C. Masouros, and J. Yuan, "Radar-Assisted Predictive Beamforming for Vehicular Links: Communication Served by Sensing," in IEEE Transactions on Wireless Communications, vol. 19, no. 11, pp. 7704-7719, Nov. 2020, doi: 10.1109/TWC.2020.3015735.
[2] Huang H, Yuan W, Liu C, et al., "LLM in V2I: A Data-Driven Predictive Beamforming Framework for Vehicle Tracking in Near-Field ISAC Systems," in IEEE Journal on Selected Areas in Communications, vol. 44, pp. 2510-2527, 2026, doi: 10.1109/JSAC.2025.3643394.

3. The background discussion does not clearly explain why conventional beamforming methods are inadequate in the considered setting. The sum-rate maximization problem in (5), under known user number, antenna number, and perfect CSI, has been extensively studied and can be addressed by many mature algorithms. The authors should more clearly specify the target deployment scenario: What latency budget is assumed? How fast does the channel vary? How often do K and N change? Under what system scale does WMMSE become infeasible? Without this clarification, readers may question why a complex DL model is necessary for solving a well-established perfect-CSI beamforming problem.

4. The current contribution section mostly describes what the authors propose. However, the gap between existing works and the proposed method is not sufficiently sharpened. For example, size generalization, masking, Transformer-based beamforming, and L2O-based beamforming have all been considered to some extent in prior works. The authors should make clearer what unique challenge is not addressed by previous methods and why the proposed design is essential.

5. Equation (5) formulates a classical sum-rate maximization problem with perfect CSI. In this setting, many non-DL and learning-assisted methods are already available. The authors should better explain the practical significance of solving this exact problem using a relatively large Transformer-based DL model.

6. The paper states that the T-layer network approximates the optimization trajectory and that each layer corresponds to an optimizer step. However, since different layers have different parameters, the proposed model can also be interpreted as a deep neural network with intermediate outputs and residual connections. The authors should clarify what distinguishes the proposed T-layer L2O model from directly training a T-layer Transformer to output the beamformer. The effect of T should also be studied more systematically. The paper fixes T=10, but does not sufficiently explain how T affects performance, inference latency, memory, and convergence.

7. The paper maximizes the cumulative sum-rate over all layers to encourage performance improvement at every optimization step. However, there is no guarantee that $R(t+1) \geq R(t)$. Thus, the authors should provide layer-wise sum-rate curves for individual samples or average trajectories, and report whether monotonicity is actually observed. If monotonic improvement is important, the authors may consider adding a monotonicity regularizer or projection/acceptance step.

8. The paper reports that the proposed scheme outperforms WMMSE in underloaded systems and approaches or surpasses WMMSE in some overloaded regimes. However, WMMSE is an iterative local optimization algorithm, and its performance depends on initialization, stopping criterion, number of iterations, and convergence tolerance. The manuscript should clearly state these settings. If WMMSE is not sufficiently converged, the comparison may overstate the advantage of the proposed method. It would also be helpful to compare against multiple WMMSE initializations or stronger optimization baselines.

9. In the sparse-channel overloaded case, the authors state that they initialize the beamformer using the result of a two-step WMMSE iteration to alleviate learning difficulty, at the cost of slower inference. This is an important caveat. It suggests that the proposed method may rely on conventional optimization assistance in difficult channel regimes. The authors should discuss whether the method remains “real-time” and whether the performance gain is due to the SALLO-M architecture itself or partly due to the stronger WMMSE-based initialization. An ablation without two-step WMMSE initialization in sparse overloaded channels would be useful.